Franciscan Eco-Theology and Environmental Attitudes

Term paper submitted for the
Research Module in Management and Applied Microeconomics

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Abstract

We analyse the effect of exposure to Franciscan and Dominican Mendicant orders between the 13th and 15th centuries on contemporary environmental attitudes. We find that higher Franciscan exposure is associated with less pro-environmental attitudes, contrary to our initial expectations, whereas the effect of Dominican exposure is statistically insignificant. We suggest that these results could be driven by differences in orientation toward collective action. This possible underlying mechanism is consistent with the contrasting theologies of the two orders regarding collectivity and individualism.

1 Introduction

The Catholic Church played an important role in medieval Europe, shaping people's lives, values, and behaviour throughout the Middle Ages (**lynch2014**). One of the most famous religious figures of the Middle Ages is St. Francis of Assisi, founder of the Franciscan Order and patron saint of ecology within the Catholic Church, widely known for his deep affection towards nature and animals.

We investigate what is the effect of exposure to the Franciscan "Eco-Theology" between the 13th and 15th centuries on contemporary environmental attitudes among Europeans. We use as a control the exposure to the Dominican order, which was another major Mendicant Order during the Middle Ages.

To do so, we construct historical exposure measures for both Franciscan and Dominican Orders as independent variables, defined as the fraction of a region's historical population living geographically close to a monastic house. We then construct our dependent variable, the environmental index, via Principal Component Analysis to individual-level survey data on environmental attitudes from the European Values Study (EVS). Using those variables and a set of individual-level controls, we empirically examine how historical exposure to the two Mendicant Orders affects contemporary environmental attitudes.

Given the historical background of St. Francis of Assisi, we have set the following initial hypothesis: higher historical exposure to the Franciscan Order is associated with more pro-environmental attitudes, while higher exposure to the Dominican Order is associated with less pro-environmental attitudes. However, contrary to our expectations, we find that higher exposure to the Franciscan Order is associated with less pro-environmental attitudes, while exposure to the Dominican Order had statistically insignificant results.

We propose a possible mechanism that might have driven those results, grounded on the underlying structural differences of both Catholic Orders on the collective action and individualism. The Franciscan theology of humility and acceptance of circumstances might have led to less pro-environmental attitudes, while the Dominican theology of persuasion and institutional coordination might have led to more pro-environmental attitudes.

2 Historical Context and Hypothesis

At a glance, both Franciscans and Dominicans were Mendicant Orders, meaning that they relied on charity and begging for their livelihood and were more committed to preaching and teaching than owning land or monastic seclusion. Founded in the same century, both orders were operating as branches of the Catholic Church and operated widespread across Europe, with significant influence on the medieval society. However, despite having these similarities, the two orders differ in their theological emphasis and approaches (**arrunada2024**).

The Franciscan Order was founded in the early 13th century by St. Francis of Assisi (1181-1226), who was recognized by the Catholic Church as the patron saint of ecology, widely known for his profound affection towards nature and animals, as seen in his "Canticles of the Creatures," where he refers to sun and moon as brother and sister. Francis, together with his followers, emphasized the close imitation of Christ and the apostles through poverty and humility. "*He wanted to embrace real poverty, without any rationalisations* *or modifications. His band of followers would beg for their daily food, would take no care for tomorrow, and would own nothing – really nothing.*" (**lynch2014**, p. 258).

In contrast, the Dominican Order, founded by St. Dominic (1170-1221) in the early 13th century, prioritized intellectual and educational mission of medieval Catholic Church. "*For Dominic, a life of poverty and begging was a means to an end, whereas for Francis, it was an end in itself*" (**lynch2014**, p. 262). Recognizing that learning was important to train preachers, the Dominican Order had internal schools (**lynch2014**). They also founded universities, and produced extensive theological and practical writings (**legoff1980**; **boyle1981**).

In short, Franciscans focus on collective life, communal care, antimaterialism, shame, compassion, simplicity, and humbleness (**arrunada2024**). In contrast, given the nature of their activities, Dominicans emphasized critical thinking, self-examination, and moral development (**hinnebusch1966**; **lesnick1989**).

Because of the strong association of the Franciscan Order with nature and animals, we hypothesized that higher historical exposure to the Franciscan Order would be associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order would be associated with less pro-environmental attitudes.

3 Literature Review

Similar to our approach, **arrunada2024** examined the effects of the two major Mendicant Orders of the Middle Ages, the Dominican and the Franciscan, on the traits of the interpersonal exchange. Through their emphasis on guilt, shame, compassion, and analytical thinking, these orders shaped cognitive, interpersonal, and institutional outcomes, which together constitute the "foundations" of impersonal exchange. They further argue that eco-theological beliefs of both Mendicant Orders "*differ sharply in their approach to both spiritual and mundane life and can foster distinct values and cultural norms*" (**arrunada2024**, p. 2).

For example, at the interpersonal level, "*Dominicans promoted guilt as a driver of prosocial behaviour toward strangers*", fostering a culture of individual accountability, independence, generalized fairness and trust, "*whereas Franciscans encouraged compassion and shame, which strengthened bonds within the group but limited cooperation beyond it*" (**arrunada2024**, p. 4).

The author's findings are consistent with our results regarding the contrasting effects of the two orders, as well as our proposed mechanism that the underlying differences in theological beliefs might have driven the results.

Our study also relates to the broader literature on the role of medieval churches and persistence of cultural norms. The Catholic Church played an influential role in shaping people's lives, values, and behaviour throughout the Middle Ages (**lynch2014**). Once a religious institution establishes, its values and norms tend to persist as cultural legacies, embedded in national cultures and are transmitted through educational institutions, even after losing some of the formal allegiance of their followers in modern Europe (**inglehart2000**).

Another important strand of literature relates to the determinants of environmental attitudes. It has been examined how cultural differences might influence environmental attitudes (**behler2025**). Some literature also suggests that religiously grounded nature world-view is associated with how people understand and relate to the environment (**cohen1976**).

On top of that, some theoretical models argue that pro-environmental attitudes may be rooted in self-interest and utility maximization. In particular, a longer life expectancy may increase individuals' willingness to support environmental efforts to fight climate change (**mariani2010**). Additionally, importantly for our research, many other studies with similar interests commonly measure the environmental attitudes using survey datasets, such as the EVS (**gelissen2007**; **behler2025**).

Taken together, these strands of literature support and motivate our research, which examines the effect of historical exposure to Franciscan and Dominican monasteries on contemporary environmental attitudes. Now the task is to come forward with a methodological strategy to isolate the effects previously discussed.

4 Data and Methodology

4.1 Exposure Measure

This study assumes that individuals residing in close proximity to a monastery were exposed to and influenced by its theological beliefs. We define geographic proximity as residence within a 25-kilometer radius of a monastery.

To establish historical monastery locations, we employ a geospatial dataset from the Mapping Past Societies (MAPS) project covering the period between 1216 and 1500. This dataset provides the geographic coordinates for 870 Dominican and 994 Franciscan monasteries.

The MAPS dataset reports foundation centuries for Dominican monasteries only, but not for Franciscan monasteries. In the absence of corresponding information for Franciscan houses, we assume that all monasteries (or their subsequent influence) persisted throughout the entire time frame. This assumption raises a key limitation, as it prevents from capturing dynamic changes in exposure, constraining the analysis to measures of aggregate exposure.

After geolocating the monasteries, we assign them to their corresponding administrative regions. For this purpose, we employ the European Commission's Nomenclature of Territorial Units for Statistics (NUTS) classification, combined with country-level shapefiles. NUTS-1 corresponds to large macro-regions, while NUTS-2 captures more localized regions within countries. Because Germany modified its NUTS-2 administrative boundaries over the EVS data collection period, we rely on NUTS-1 regions for Germany and NUTS-2 regions for all other countries.

We exclude monasteries located substantially beyond the relevant NUTS regions, such as those in Eastern Ukraine, Crimea, and the Levant. Monasteries situated just outside formal NUTS boundaries but geographically adjacent to a NUTS region are assigned to the nearest valid region, as illustrated in Figure 1.

The dataset includes some monasteries located in Turkey, as the EU also collects data for parts of Turkish territory. However, these observations are not included in the analysis due to the absence of corresponding environmental attitudes data. This does not affect the results or procedure. Similarly, some monasteries are also observed in Greece, but these do not affect the quantitative approach, since there are no entries for Greece in the EVS environmental survey data.

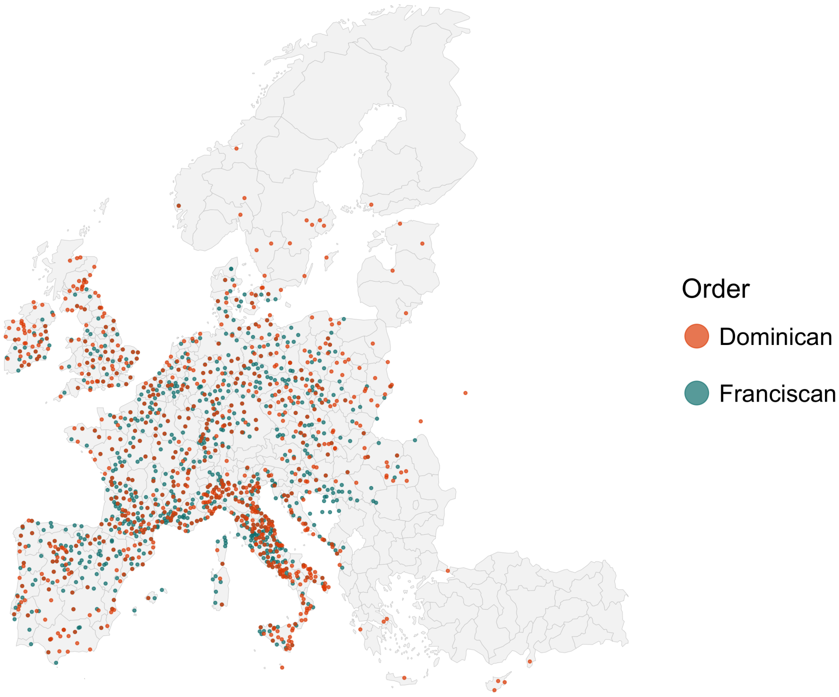


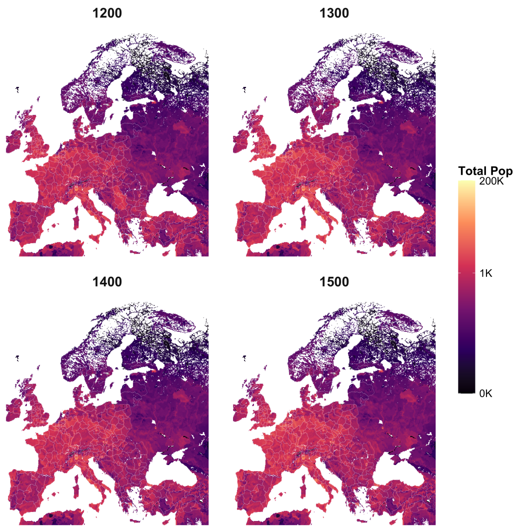
Figure 1: Locations of Franciscan and Dominican Monasteries

Once we have combined the monastery locations with NUTS shapefiles, the next step is to observe historical population data. To this end, we employ the HYDE 3.3 (History Database of the Global Environment) dataset maintained by Utrecht University, which captures population numbers historically. Grid cells are squares of 85 km^2, which is relatively granular, and population estimates are available for each grid. In Figure 2a, each cell represents estimated population counts for a historical year (1200–1500).

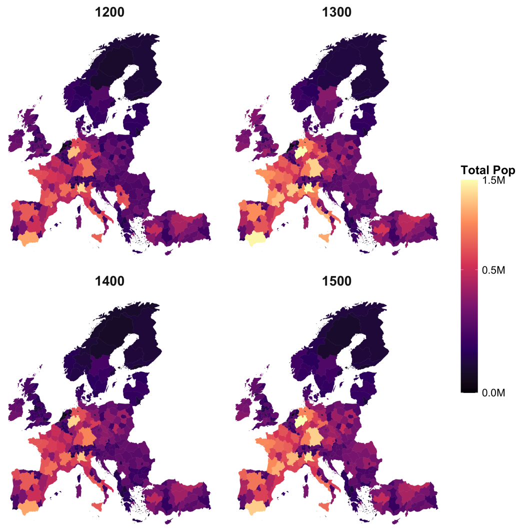
Based on the population grid, we construct estimates at the NUTS level. In Figure 2b, population data from HYDE 3.3 grids are overlaid on NUTS boundaries, with total population calculated as the sum of the population of all grid cells that fall within. If a cell is partially within a NUTS region, it counts as a full cell if more than half of its area falls within the region. This way, we can estimate the population of each NUTS region based on the HYDE grid data.

Finally, we match the population estimates with the monastery locations. As previously stated, we define the exposed population using a 25-kilometer radius around each monastery. Grid cells with more than half of their area falling within this radius are classified as fully exposed to the respective order's theology.

$$\text{Exposure} = \frac{\text{Exposed Population}}{\text{Total Population}} \tag{1}$$



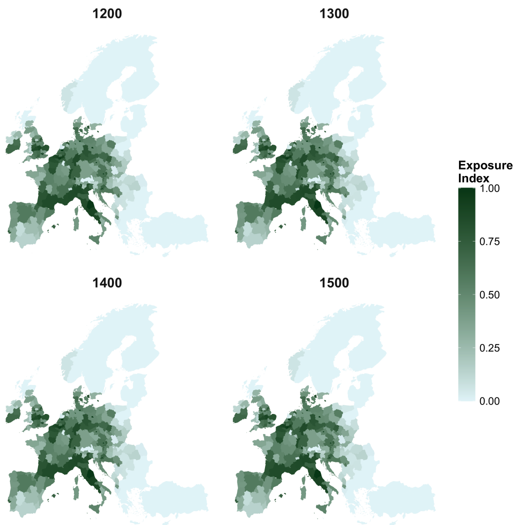
(a) Population densities (grid-level, log scale)



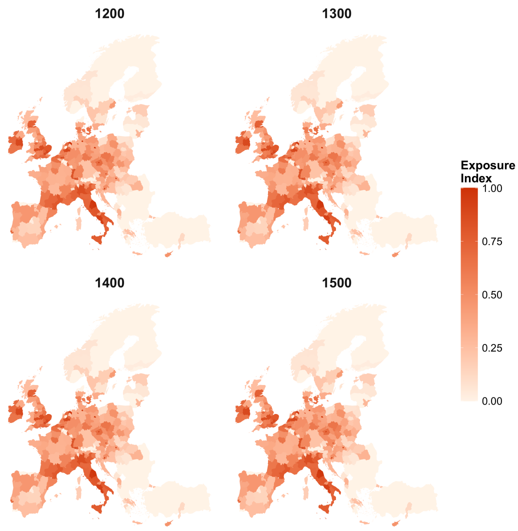
(b) Regional populations (NUTS-2 level)

Figure 2: Population measures from HYDE 3.3

With all three data sources combined, we have Dominican and Franciscan exposures:



(a) Franciscan exposure



(b) Dominican exposure

Figure 3: Map of exposure to Mendicant Orders in Europe (1200–1500)
normalized index (0 = none, 1 = high exposure)

4.2 Environmental Attitudes Measure

To construct the main dependent variable we use the European Values Study (EVS) 2017–2021 dataset, which contains individual-level survey data on values, beliefs, attitudes, and socioeconomic characteristics across European countries. The dataset covers over 30 European countries, 24 of which are relevant for our research. After data cleaning, the final dataset includes 49233 observations and geographic identifiers for NUTS-1 regions for Germany and NUTS-2 for all the other countries.

From the EVS dataset, we select six environmental variables in particular, which correspond to questions related to environmental attitudes. The questions are the following:

Survey Question (EVS)	**Dimension**
Choosing between "protecting the environment priority, even if slower economic growth and loss of jobs" vs. "economic growth and creating jobs priority, even if environment suffers"	Economy–environment trade-off
"I would give part of my income if I were certain that the money would be used to prevent environmental pollution."	Willingness to pay
"It is just too difficult for someone like me to do much about the environment."	Self-efficacy
"There are more important things to do in life than protect the environment."	Priority of environmental protection
"There is no point in doing what I can for the environment unless others do the same."	Collective action
"Many of the claims about environmental threats are exaggerated."	Env. skepticism

The questions have categorical answers ("agree strongly", "agree", "neither agree nor disagree", "disagree", "disagree strongly", or choosing between the two statements). However, the questions differ in their directional interpretation. Thus, we need to standardize the questions by assigning a 0-1 scale for each answer to each question (see Table 2), such that the answers indicate the same environmental direction.

In the standardized 0-1 scale, 1 indicates the least pro-environmental attitude, and 0 indicates the most pro-environmental attitude. See Figure 4 for the visualization for environmental questions within the NUTS regions. The green regions indicate being more pro-environmental, while the yellow regions indicate less pro-environmental attitudes.

Since we have 6 different closely related questions for environmental attitudes, we construct a single Environmental Attitudes Index. To do so, we run a Principal Component Analysis (PCA) to combine all 6 variables. The first principal component explains 22.8% of the variance, and we take the first principal component as our Environmental Attitudes Index (see Table 3 in the Appendix for PC1 loadings). The index is continuous between 0 and 1, with 0 being more pro-environment, and 1 being less pro-environment.

We find that the components of Western European countries behave more pro-environment (green), while generally Eastern European countries have less pro-environmental attitudes (yellow), as illustrated in Figure 5.

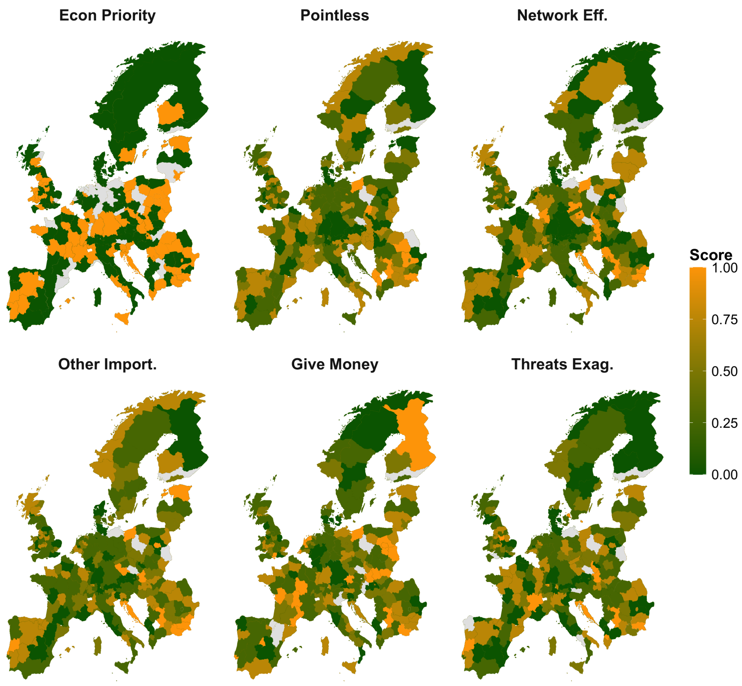


Figure 4: Distribution of the 6 environmental questions across Europe considering the average scores per region.

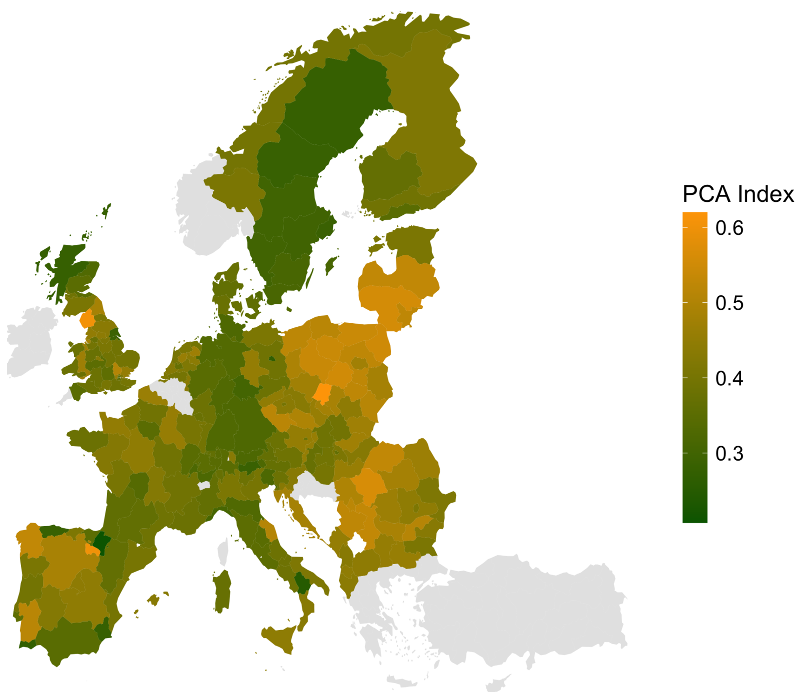


Figure 5: Environmental Index (PCA) across Europe considering the average scores per region.

5 Analysis and Results

5.1 Main Specification

Once we have all the variables together, we start our regression analysis. Our main specification is:

$$Y_{irc} = \alpha + \beta_1 Franc_r + \beta_2 Dom_r + X_i'\gamma + \delta_c + \varepsilon_{irc} \tag{2}$$

where Y_{irc} denotes the environmental attitude index for individual i in region r of country c (scaled 0-1), $Franc_r$ and Dom_r represent population-weighted exposure to Franciscan and Dominican Orders in region r measured in 1500, and X_i' is a vector of control variables, δ_c denotes country fixed effects, and ε_{irc} is the error term.

The control variables were taken from the EVS data. The vector of individual-level control variables, X_i', consists of: gender, age, education level, household income (PPP-adjusted), socio-economic status (ISEI), town size, political orientation (left–right scale), and religious denomination with dummy variables for Catholic and Protestant. We cluster standard errors at the NUTS level.

5.2 Regression Results

Table 1 presents the baseline regression results. Model 1 regresses the environmental index solely on Franciscan exposure, yielding a negative and statistically significant coefficient (β_1). Because the index is scaled from 0 (most pro-environmental) to 1 (least pro-environmental), this negative coefficient indicates a positive correlation with pro-environmental attitudes, aligning with our initial hypothesis. In Model 2, the inclusion of Dominican exposure attenuates the magnitude of β_1. When country fixed effects are introduced in Model 3, β_1 changes sign and loses statistical significance.

In our full specification (Model 4), which incorporates all individual-level controls, Dominican exposure, and country fixed effects, β_1 becomes positive and statistically significant. This indicates that moving from zero to maximum Franciscan exposure is associated with a 3.5 percentage point increase in the environmental index (i.e., less pro-environmental attitudes), directly contradicting our initial hypothesis. Notably, several control variables are also significant; for instance, identifying as female correlates with more pro-environmental attitudes. The coefficient for Dominican exposure (β_2) remains statistically insignificant across all specifications.

All the β_1 and β_2 coefficients from these regressions are plotted in Figure 6. β_1 has a steady increase going from a negative to a positive sign as we move from Model 1 to Model 4, and finally becomes positive and statistically significant. β_2 is never significant.

5.3 Robustness Checks

To assess whether a specific component of the PCA index drives these results, Table 4 presents regressions for each environmental question separately using the full specification (Model 4). While the magnitudes of β_1 and β_2 remain relatively stable across these disaggregated models, the only statistically significant effects emerge in the context of collective action ("There is no point in doing what I can for the environment unless others do the same"). The Franciscan and Dominican exposure coefficients yield contrasting,

Table 1: Regression Results (SE clustered by region)

	Model 1 Franciscans	Model 2 Fran vs Dom	Model 3 Country FE	Model 4 Full + FE
Franciscan Exposure	-0.095^{***}	-0.058^{*}	0.004	0.035^{*}
	(0.016)	(0.027)	(0.014)	(0.015)
Dominican Exposure	–	-0.050	-0.013	-0.016
		(0.027)	(0.014)	(0.014)
Female	–	–	–	-0.028^{***}
				(0.003)
Age	–	–	–	0.000^{**}
				(0.000)
Education	–	–	–	-0.000^{***}
				(0.000)
Income (PPP)	–	–	–	-0.002
				(0.001)
Right-wing	–	–	–	0.007^{***}
				(0.001)
Town Size	–	–	–	-0.005^{**}
				(0.001)
Socio-Economic Index	–	–	–	-0.001^{***}
				(0.000)
Catholic	–	–	–	0.004
				(0.004)
Protestant	–	–	–	-0.001
				(0.005)
Observations	49,233	49,233	49,233	24,379
R^2	0.022	0.024	0.099	0.149
Adj. R^2	0.022	0.024	0.099	0.148

Notes: Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

significant results for this specific variable. This divergence is consistent with the theological approaches of the two orders, where the Franciscans placed greater emphasis on collective, communal living, while the Dominicans favored an individualistic framework.

To ensure our results are not driven by outlier countries, Figure 7 presents a leave-one-out sensitivity analysis estimating Model 4 while iteratively excluding one country at a time. The top and bottom plots display the estimated coefficients for Dominican (β_2) and Franciscan (β_1) exposure, respectively. The estimates remain highly stable, demonstrating that the observed effects are homogeneous across Europe. Only the exclusion of Switzerland or Italy yields minor deviations, though the core findings remain consistent with Table 1.

In Table 5, we run the main regression again, this time excluding each control variable at a time for the sensitivity analysis. We see that the estimate of β_1 remains around 0.035 and is statistically significant across most of the regressions, except for one regression in which we exclude the town size control variable. This is an important result, as the Franciscans choose to locate themselves within urban areas to reach more people. Thus, not controlling for the town size would bias the coefficient of Franciscan exposure. This makes the town size a very important control, but apart from it, we could drop any other control and find very similar results to our full model.

In Figure 8, we see the plotted regression coefficient estimates found in the sensitivity analysis in Table 5 by leaving out one control in each regression, the top plot showing estimates for β_2 and the bottom one showing estimates of β_1. We can clearly see that only the town size shifts the Franciscan exposure coefficient estimate. The Dominican exposure coefficient estimate also changes slightly when we remove income, political orientation, and town size, but none of the controls change the general direction of the results.

As additional robustness checks in Table 6, we run subsample regressions with the full model, including all controls, but keeping only Catholic individuals in the first subsample. The Catholicism of individuals is determined from the same EVS data by asking the question "Which religious denomination do you belong to?" and taking into account only those who answered with the option "Roman Catholic". Considering only Catholics, the sample size drops significantly to 5899, and we don't have any significant results. In the second subsample, we exclude the observations from Germany to only keep NUTS-2 regions, to account for the lack of granularity in Germany. However, the results stay the same as in the main model.

Lastly, we extend the analysis by running the same four models as in Table 1 but with the same restricted sample size that we have in Model 4 (full specification). When we include all controls, the number of observations falls significantly from a full sample size of 49233 to 24379 observations. We lose a large number of observations because of the missing values for the control variables. When someone did not provide an answer to a question in the survey that we use as a control, we exclude that observation from the sample. All remaining observations are such that there are answers to all control variables. However, this allows us to check whether the results and the change in sign are driven by the sample composition. To rule out potential selection bias arising from missing observations in the survey control variables, we re-estimate all four models on a restricted sample containing no missing covariates in Figure 9. The results remain fully consistent with our baseline findings.

5.4 Possible Mechanism

Since the results differ from our prior expectations, we find it important to propose a possible mechanism that might be driving these results, for further research.

Our set of individual-level controls and country fixed effects helps mitigate several confounding channels. For instance, the (insignificant) positive environmental attitudes associated with Dominican exposure are not simply a byproduct of higher regional education levels driven by the order's historical emphasis on teaching, as education is controlled for in our full specification. Therefore, the divergence in contemporary attitudes is likely rooted in deep, slow-moving cultural or institutional traits.

We propose that the mechanism could lie in the opposite beliefs about taking collective action within the Franciscan versus Dominican theologies. The Franciscan theology prioritizes humility, acceptance of circumstances, poverty, and scarcity for the sake of imitating Christ, thus valuing moral purity over institutional action. The lack of desire for collective intervention might have shown itself through having less pro-environmental attitudes, which could also be observed in our Table 4 results (specifically beliefs about individual efforts being pointless unless others do the same, and environmental threats being exaggerated).

We theorize that the Franciscan legacy is not inherently anti-nature, but rather anti-interventionist. On the other hand, Dominican theology highlights persuasion and institutional coordination. This willingness to enforce norms and engage in reasoned collective action may have translated into stronger contemporary trust in institutional solutions to environmental threats.

6 Limitations and Further Research

One main limitation of our research is the Mapping Past Societies (MAPS) dataset, which contains foundation centuries only for Dominican monasteries, but not the specific dates, and it doesn't contain closing dates for any monasteries. Thus, we had to assume that all the monasteries were active throughout the period of 1200-1500, and so our exposure measure treats monastery presence as time-invariant.

The main extension that we propose is to use data on the foundation and closing dates of the monasteries to construct a more accurate exposure measure. We propose to use another existing dataset from Boranbay and Guerriero (2019) specifically for the Franciscan monasteries. They use a novel dataset which contains 2980 observations (while only 994 observations in MAPS) with specific foundation dates and monastery names for Franciscan monasteries. However, there are still no closing dates available in this dataset. Still, this would allow us to construct a more accurate exposure measure for the Franciscan monasteries.

Furthermore, our empirical strategy is purely historical and correlational. Because medieval monastery locations were not randomly assigned, we cannot definitively rule out unobserved regional characteristics that might have influenced both monastic settlement and modern cultural attitudes. Future research could address this potential endogeneity by employing instrumental variable (IV) strategies—such as distance to historical trade routes—or by explicitly modeling endogenous monastery placement to improve causal identification. There is also a problem with the fact that there are no Franciscan monasteries in some NUTS regions in Northern Europe, or no Dominican monasteries in some NUTS regions in Eastern Europe. Therefore, we cannot compare the effects of the two orders in those regions. A possible extension is thus to exclude the regions that do not have either of the monasteries.

A related limitation stems from our dependency on historical exposure from more than five centuries ago, and assumption that this influence has persisted within local populations. However, modern environmental attitudes measured in the EVS may not perfectly reflect the exposure of current residents due to migration flows. Future studies should account for internal and cross-border migration to refine these regional exposure estimates.

Finally, while our covariates successfully control for many individual-level traits, there is a risk of over-controlling (i.e., including "bad controls"). Variables such as income, education, and political orientation might themselves be outcomes of historical monastic exposure. Including them in the regression could absorb part of the true historical effect. A future research extention could thus be to consider more carefully the control variables.

As other possible extensions, we could check for the choice of monastery locations. It is possible (and expected) that monasteries were not randomly located across Europe. In this scenario, we could construct instrumental variables for the monastery locations, such as the distance to the nearest city or trade route, use additional controls, or explicitly model the endogenous placement to improve the causal identification and remove the endogeneity problem.

We could also construct a distance-weighted or decay-based exposure measure. Instead of assuming that everyone within a 25 km radius of the monastery is exposed to the theology of that monastery in the same way, we could assume that the influence decays with distance and people are less exposed the further they are.

We would also like to test the effects of monasteries on other outcomes, such as climate policy support, trust in institutions, and collective action. By using other datasets, we could extend our research to other countries outside of Europe, such as Latin America, where the influence of the Catholic Church is also quite strong. For example, **waldinger2017** uses a Mexican dataset to show the long-term effects of missionary orders, such as the Franciscans, Dominicans, and the Jesuits, on outcomes such as education and Catholicism.

In summary, investigating long-run cultural persistence using historical data inherently requires strong identifying assumptions. It is fundamental to evaluate our empirical choices against an ideal experimental benchmark, such as the random historical assignment of monasteries, to explicitly recognize where data constraints force deviations from ideal causal inference. Acknowledging these limitations ensures that our results are interpreted with appropriate nuance. Ultimately, we hope this research provides a robust baseline for future studies exploring the long-term institutional and theological determinants of modern environmental attitudes.

7 Conclusion

In this research, we analyzed whether the exposure to the two major Mendicant Orders of the Middle Ages, Franciscan and Dominican monasteries, has an effect on contemporary environmental attitudes. We constructed an exposure measure by combining the geo-dataset of monastery locations with historical population data, and we constructed an environmental attitudes index using the European Values Study survey data on individual environmental questions. We ran regressions controlling for various individual-level characteristics and country fixed effects.

Through this, we have found that higher Franciscan exposure is associated with less pro-environmental attitudes. More specifically, in our main regression model, β_1 is 0.035 and statistically significant. Through various robustness checks, we have found that the results are quite robust and homogeneous across Europe and persist even when we exclude certain controls or countries.

We also proposed a possible mechanism that could be driving these results, which is the contrasting beliefs about collective action within the Franciscan and Dominican theologies. The Franciscan theology of humility and acceptance of circumstances might have led to less pro-environmental attitudes, while the Dominican theology of persuasion and institutional coordination might have led to more pro-environmental attitudes.

Thus, the results might have been driven by underlying beliefs about whether environmental threats are exaggerated and whether individual efforts are pointless unless others do the same for the environment.

We can conclude our results by saying that long-run cultural and theological beliefs and monasteries can have a significant influence on contemporary attitudes toward the environment. The results of this research are quite interesting and surprising, but we cannot help but mention the limitations of our research and the need for further research.

Appendix I: Environmental Attitude

Table 2 reports the exact wording and coding of the six environmental attitude questions from the EVS used in the construction of the Environmental Attitudes Index. Five of the questions are originally measured on a five-point Likert scale. One question gave the possibility to choose between two statements. In this case, choosing protecting the environment is coded as 1, and choosing economic growth is coded as 5.

We rescale all variables to an interval between 0 and 1 and treat them as continuous in the analysis to make it easy to combine all questions into our PCA on a 0-1 scale. Due to the discrete nature of the original response categories, the rescaled variables only assume the values 0.0, 0.2, 0.4, 0.6, 0.8, and 1.0. Higher values indicate less pro-environmental attitudes, while lower values indicate more pro-environmental attitudes. The rescaling is implemented using a linear transformation of each variable to the unit interval.

Table 2: Environmental Attitude Survey Questions (European Values Study)

Survey Question (EVS)	**Scale**
Choosing between "protecting the environment priority, even if slower economic growth and loss of jobs" vs. "economic growth and creating jobs priority, even if environment suffers"	only 1 and 5 to [0,1]
"I would give part of my income if I were certain that the money would be used to prevent environmental pollution."	1, 2, 3, 4, 5 to [0,1]
"It is just too difficult for someone like me to do much about the environment."	1, 2, 3, 4, 5 to [0,1]
"There are more important things to do in life than protect the environment."	1, 2, 3, 4, 5 to [0,1]
"There is no point in doing what I can for the environment unless others do the same."	1, 2, 3, 4, 5 to [0,1]
"Many of the claims about environmental threats are exaggerated."	1, 2, 3, 4, 5 to [0,1]

Table 3: Environmental Index: PC1 Loadings

	PC1 Loading
envir_econ_priority	0.855
envir_efforts_pointless	-0.287
envir_other_importances	0.498
envir_network_effect	0.200
envir_threats_exaggerated	0.510
envir_protection_money	0.085
	Value
SS Loadings (PC1)	1.368
Proportion of Variance Explained	0.228

Appendix II: Regression Results and Robustness Checks

Figure 6: Impact of Monastic Orders (Clustered SEs)

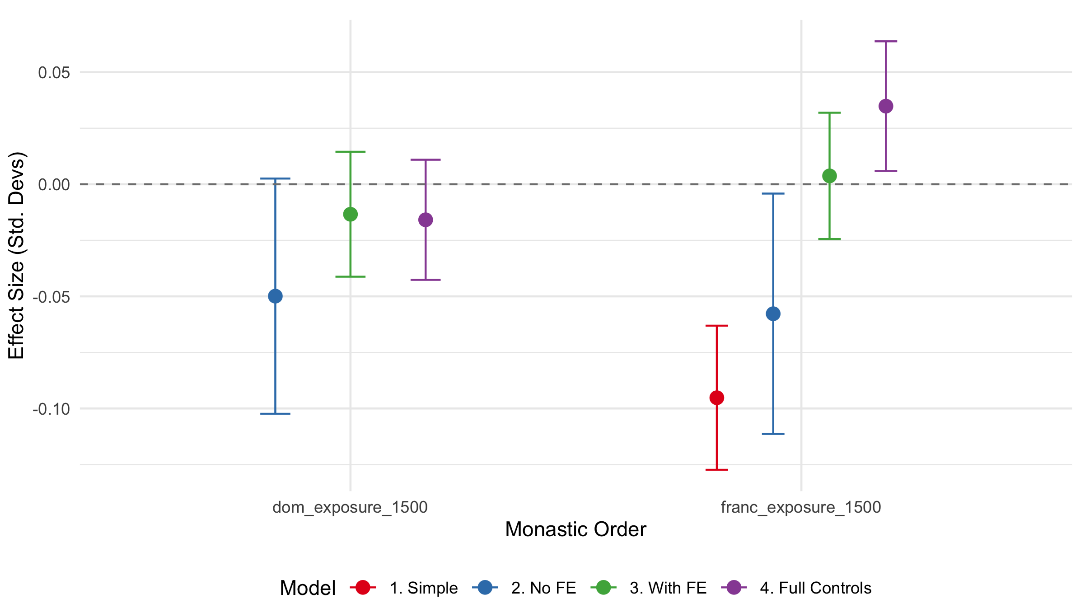


Table 4: Individual Environmental Questions (Clustered SEs)

	(1) Econ Priority	(2) Give Money	(3) Too difficult	(4) Other Import.	(5) Collective	(6) Threats Exag.
Franciscan Exp.	0.052	0.016	0.034	0.030	0.064**	0.037
	(0.038)	(0.021)	(0.021)	(0.019)	(0.021)	(0.021)
Dominican Exp.	-0.020	-0.026	-0.014	-0.018	-0.039*	-0.017
	(0.037)	(0.021)	(0.018)	(0.016)	(0.016)	(0.015)
Gender (Female)	-0.027***	-0.002	-0.023***	-0.040***	-0.028***	-0.047***
	(0.007)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)
Age	0.001**	0.001***	0.002***	0.000*	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Education	-0.000**	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Income (PPP)	-0.002	-0.004*	-0.011***	-0.004**	-0.010***	-0.004**
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
Socio-Econ. Ind.	-0.002***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Town Size	-0.003	-0.006**	-0.001	-0.004*	-0.009***	-0.012***
	(0.004)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Polit. Orient. (R)	0.011***	0.006***	0.003**	0.004***	0.005***	0.010***
	(0.003)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Catholic	0.007	0.005	0.011*	0.010	0.017*	0.007
	(0.009)	(0.007)	(0.006)	(0.005)	(0.007)	(0.007)
Protestant	-0.003	0.003	-0.007	-0.017**	-0.004	0.008
	(0.013)	(0.012)	(0.006)	(0.005)	(0.006)	(0.008)
Num. Obs.	25,427	27,212	27,344	27,370	27,343	26,823
R^2	0.095	0.121	0.140	0.141	0.125	0.121
Adj. R^2	0.094	0.120	0.139	0.139	0.124	0.120

Notes: Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 7: Coefficients when excluding each Country

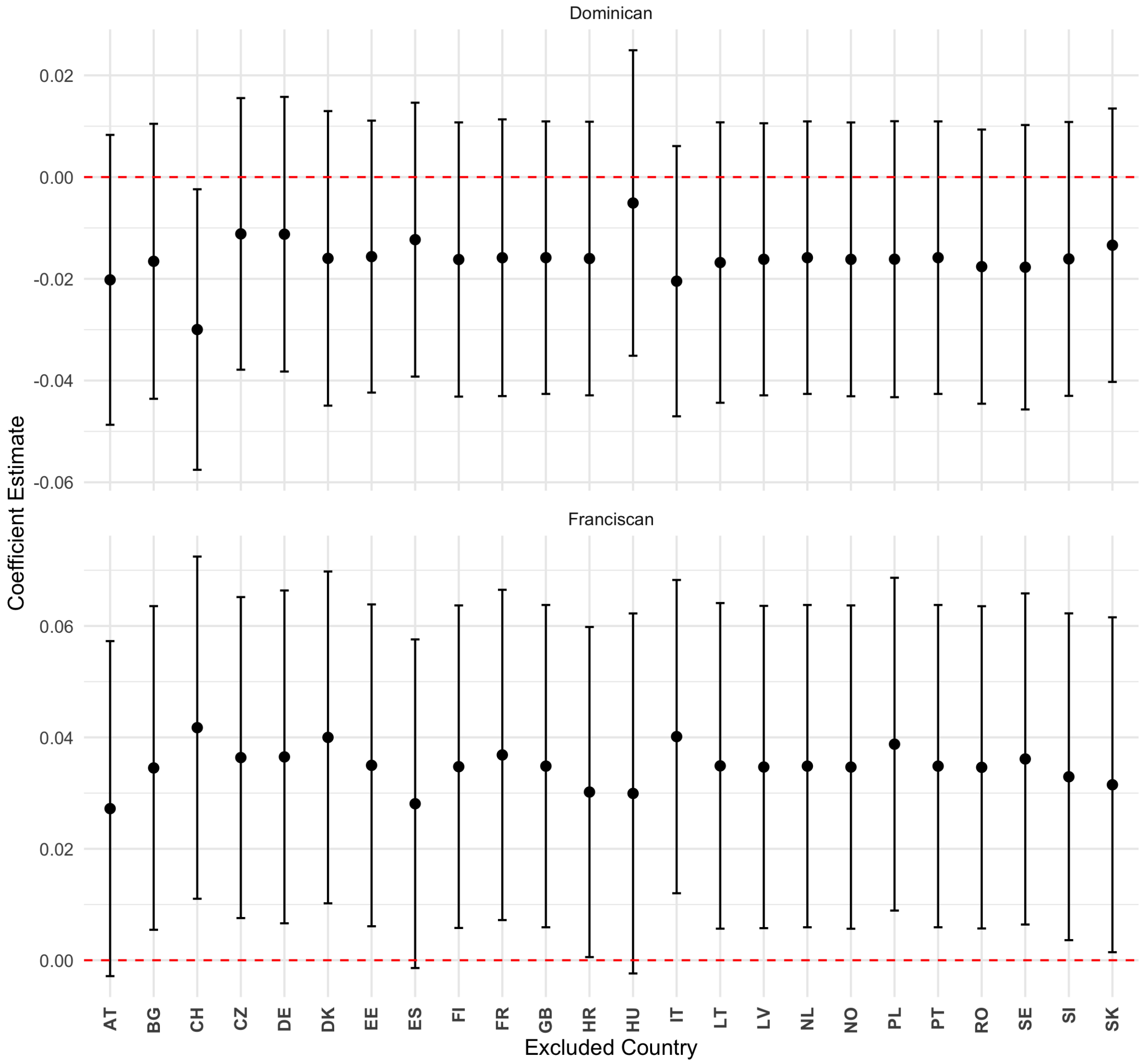


Table 5: Sensitivity Analysis: Coefficients when Dropping Controls

	Full Model	Excl. Gender	Excl. Age	Excl. Education	Excl. Income	Excl. Politics	Excl. Town Size	Excl. Socio-Econ. Index	Excl. Religion
Franc. Exp.	0.035*	0.036*	0.035*	0.035*	0.032*	0.033*	0.012	0.039**	0.034*
Dom. Exp.	-0.016	-0.016	-0.015	-0.017	-0.008	-0.010	-0.010	-0.020	-0.016
Num. Obs.	24,379	24,379	24,434	24,438	26,795	33,560	28,220	27,286	24,379
R^2	0.149	0.144	0.149	0.147	0.143	0.135	0.141	0.148	0.149
Adj. R^2	0.148	0.143	0.147	0.145	0.142	0.134	0.139	0.147	0.148

Notes: $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 8: Sensitivity Analysis Plot

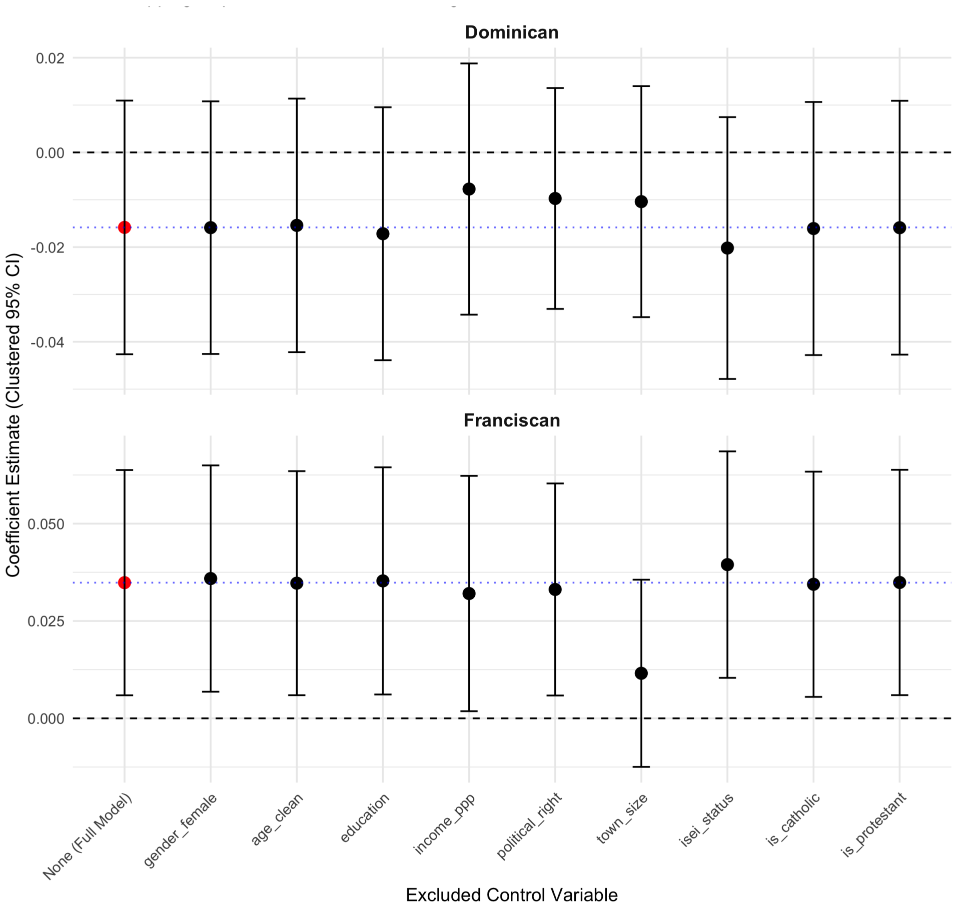
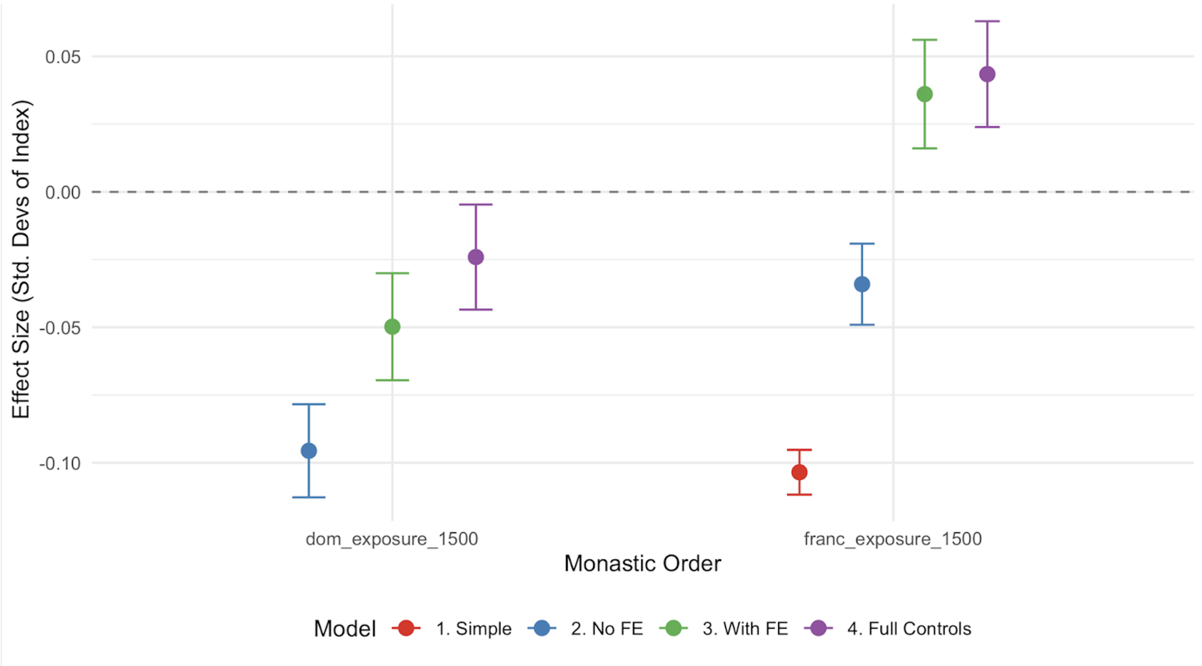


Table 6: Robustness Checks: Subsamples

	Main Model (All)	Catholics Only	Exclude Germany
Franciscan Exposure	0.035*	0.023	0.037*
	(0.015)	(0.021)	(0.015)
Dominican Exposure	-0.016	0.004	-0.011
	(0.014)	(0.018)	(0.014)
Num. Obs.	24,379	5,899	23,182
R^2	0.149	0.132	0.146
Adj. R^2	0.148	0.126	0.145

Notes: Standard errors in parentheses. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 9: Regression models of Table 1, using the same sample



Appendix III: The ChatGPT 5.2 Prompts

In this paper, we consulted ChatGPT for specific editing-related tasks, including identifying errors in code segments that were not running as intended, assisting with LaTeX formatting, and improving grammar and flow. The prompts did not always yield ready-to-use results. We therefore carefully reviewed and verified all outputs, as we were aware that ChatGPT could produce incorrect or misleading suggestions. The specific prompts used are listed below.

Code Debugging for Data Preparation and Analysis

> Please help identify the problem in the following lines of code:
> *[Area where specific code lines and the corresponding error message was provided.]*

This prompt was used to debug code segments that did not initially work as intended during the construction of the exposure measure, the environmental attitudes index, and the generation of plots and regression results.

LaTeX Formatting

> Please typeset the following equation/line in LaTeX:
> *[Area where the text or equation was provided.]*
>
> There is an error in the LaTeX equation. Please help to solve the error:
>
> ```
> \begin{equation} Y_{irc} = \alpha + \beta_1 Franc_{r} + \beta_2 Dom_{r}
>  + X_i' \gamma + \delta_{c} + \varepsilon_{irc} \end{equation}
> ```
>
> Please transform our regression results *[screenshot of the regression table]* into a shorter and cleaner table, such that the numbers are larger and more readable for presentation.
>
> How to link code-generated plots into LaTeX?
>
> The plots take up substantial space in our paper, and we are constrained by a page limit. How to arrange these plots horizontally and adjust their size? *[Area where existing code lines for plotting were provided.]*
>
> Please typeset the following citation in LaTeX format for the bibliography section of the paper: *[Area where the APA-formatted citation was provided.]*

These and similar formatting-related prompts were used whenever we needed assistance converting specific elements into LaTeX format.

Language Polishing

> Please review the following text for grammar, clarity, and flow. Revise for readability while preserving the original meaning and academic tone. *[Area where the text was placed. Short excerpts (usually no more than 2–3 sentences at a time) from selected parts of the paper were provided.]*

This prompt was used consistently during the editing process to improve the clarity and readability of selected sections of the text.